Example 2.1 We have 34 cakes. Each box can hold 5 cakes only. How many boxes we need to pack and how many cakes are unpacked?

Solution We see that 6 boxes are required to pack 30 cakes with 4 cakes left over. This distribution of cakes can be understood as follows:

34	=	5	×	6	+	4
Total number of cakes	=	Number of cakes in each box	×	Number of boxes	+	Number of cakes left over
↓		↓		↓		↓
(Dividend) a	=	(Divisor) b	×	(Quotient) q	+	(Remainder) r

Note

- The above lemma is nothing but a restatement of the long division process, the integers q and r are called quotient and remainder respectively.
- When a positive integer is divided by 2 the remainder is either 0 or 1. So, any positive integer will of the form $2k$, $2k+1$ for some integer k.

Euclid's Division Lemma can be generalised to any two integers.

Generalised form of Euclid's division lemma

If a and b are $(b \neq 0)$ any two integers then there exist unique integers q and r such that $a = bq + r$, where $0 \leq r < |b|$

Example 2.2 Find the quotient and remainder when a is divided by b in the following cases (i) $a = -12,\ b = 5$ (ii) $a = 17,\ b = -3$ (iii) $a = -19,\ b = -4$

Solutions

(i) $a = -12,\ b = 5$

By Euclid's division lemma

$a = bq + r$, where $0 \leq r < |b|$

$-12 = 5 \times (-3) + 3 \quad 0 \leq r < |5|$

Therefore, Quotient $q = -3$, Remainder $r = 3$

Thinking Corner

When a positive integer is divided by 3

1. What are the possible remainders?
2. In which form can it be written?

(ii) $a = 17 \quad b = -3$

By Euclid's division lemma

$a = bq + r$, where $0 \leq r < |b|$

$17 = (-3) \times (-5) + 2,\ 0 \leq r < |-3|$

Therefore Quotient $q = -5$,

Remainder $r = 2$

(iii) $a = -19,\ b = -4$

By Euclid's division lemma

$a = bq + r$, where $0 \leq r < |b|$

$-19 = (-4) \times (5) + 1,\ 0 \leq r < |-4|$

Therefore Quotient $q = 5$, Remainder $r = 1$.

Progress Check

Find q and r for the following pairs of integers a and b satisfying $a = bq + r$.

1. $a = 13,\ \ b = 3$
2. $a = 18,\ \ b = 4$
3. $a = 21,\ \ b = -4$
4. $a = -32,\ b = -12$
5. $a = -31,\ b = 7$

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